FoodLogAthl-218: Constructing a Real-World Food Image Dataset Using Dietary Management Applications

Taro Yamada

Abstract

Food recognition models for dietary logging are typically trained on web-sourced images that differ significantly from real-world personal meal photos. We present FoodLogAthl-218, a novel food image dataset constructed from actual user logs in a dietary management application, preserving natural class imbalance, casual imaging conditions, multi-dish meals, and temporal continuity. The dataset contains 6,925 images with 14,349 dish bounding boxes across 218 consolidated classes, derived from 11,460 initial images collected from 632 users over 18 months. We introduce three evaluation tasks: (1) closed-set dish classification, (2) continual user-aware adaptation over time, and (3) menu-context-aware classification in multi-dish images. Comprehensive experiments with large multimodal models (LMMs) and CLIP variants reveal that FoodLogAthl-218 presents significantly greater challenges than existing datasets, with GPT-4o achieving only 42.4% top-1 accuracy. Our analysis demonstrates substantial benefits from continual adaptation (>30 percentage point improvements) and personalization, while menu-aware classification shows precision-recall trade-offs. FoodLogAthl-218 provides a realistic benchmark that exposes gaps in current models and enables research on practical food recognition challenges.

1 Introduction

Food image recognition has become increasingly important for dietary logging and health monitoring applications [1, 8]. However, most existing food recognition models are trained on carefully curated datasets sourced from the web [2, 11], which often feature well-lit, aesthetically pleasing images that differ substantially from the casual photos users typically capture during real-world food logging.

This domain gap presents significant challenges for practical deployment. Web-sourced datasets like Food-101 [2] and UEC-Food 256 [8] contain professionally photographed or carefully selected images with balanced class distributions. In contrast, real-world food logging involves varied lighting conditions, natural class imbalances reflecting actual eating patterns, multi-dish compositions, and temporal patterns in user behavior.

To address this gap, we introduce **FoodLogAthl-218**, a novel dataset constructed bottom-up from actual user submissions in a dietary management application. Unlike traditional top-down dataset construction that starts with predefined food categories and sources representative images, our approach begins with real user logs and then consolidates labels afterward, preserving the natural distribution and challenges of real-world food recognition.

Our key contributions are:

(1) A public, ethically-vetted real-world food image dataset containing 6,925 images with 14,349 bounding boxes across 218 classes, constructed from actual dietary app usage with rich temporal metadata.
(2) A reproducible data curation pipeline combining automated CLIP-based outlier detection, GPT-4o label consolidation, and manual quality control to handle the inherent noise in user-generated content.
(3) Three comprehensive benchmark tasks designed to capture practical deployment challenges: closed-set classification, continual user-aware adaptation, and context-aware multi-dish classification.
(4) Extensive evaluation with state-of-the-art large multimodal models and vision-language models, demonstrating significantly greater difficulty compared to existing datasets and substantial benefits from personalization and temporal adaptation.

2 Related Work

Food Image Datasets. Existing food image datasets can be broadly categorized by their data sources and construction methodologies. Early datasets like PFID [3] focused on fast food chains with limited diversity. The seminal Food-101 [2] dataset established the benchmark with 101 classes and 101,000 images sourced from foodspotting.com, emphasizing visual quality and balanced class distribution. UEC-Food 256 [8] expanded the scope with 256 categories and international cuisine, combining web sources with original databases.

Recent large-scale efforts include Food2K [11] with over 1 million images across 2,000 classes from commercial platforms, and Food-500 Cap [9] incorporating detailed captions for 494 categories. However, these datasets share a common limitation: they prioritize visual appeal and balanced representations rather than reflecting real-world usage patterns [12].

Table 3 provides a comprehensive comparison of major food image datasets. Notably, all existing large-scale datasets rely on web sources, resulting in curated, aesthetically pleasing images that may not generalize to casual user photography in dietary applications.

Real-World Food Recognition Challenges. Recent work has highlighted the gap between laboratory benchmarks and real-world performance in food recognition. Challenges include natural class imbalance, varied imaging conditions, temporal patterns in eating behavior, and the need for personalized models. However, most studies address these issues through data augmentation or algorithmic improvements rather than constructing datasets that inherently reflect these challenges.

Continual Learning in Vision. Our work connects to continual learning research, particularly in computer vision applications where models must adapt to new data distributions over time. While most continual learning work focuses on catastrophic forgetting, our temporal adaptation task examines how models can improve performance through exposure to user-specific patterns over time.

FoodLogAthl-218 distinguishes itself by being the first large-scale food dataset constructed directly from real dietary management app usage, preserving the natural challenges and temporal patterns inherent in practical deployment scenarios.

3 FoodLogAthl-218 Dataset

3.1 Data Collection and Sources

FoodLogAthl-218 is constructed from user logs collected through the FoodLog Athl dietary management application over an 18-month period from May 2023 to October 2024. The application allows users to photograph their meals, automatically detect dishes using bounding boxes, and assign food labels from a comprehensive database containing over 140,000 items. Users can subsequently correct both the bounding boxes and labels, providing valuable ground-truth annotations for real-world food recognition scenarios.

The raw dataset initially contained 11,460 images with 61,414 bounding boxes and 5,889 distinct labels from 632 users. Each record included anonymized user ID, timestamp, dish names, portion sizes, and initial automatic detections. All data collection was conducted under appropriate ethics approval with informed user consent, and all personally identifiable information was removed through a two-stage manual review process.

3.2 Data Construction Pipeline

We developed a multi-stage curation pipeline to transform the noisy user-generated data into a high-quality benchmark dataset, as summarized in Table 1.

Step 1-2: Basic Filtering. We removed samples without valid images or bounding boxes, reducing the dataset to 11,285 images with 26,065 bounding boxes across 3,071 classes.

Step 3: Class Frequency Filtering. To ensure sufficient samples for meaningful evaluation, we removed classes with fewer than 10 instances, resulting in 9,121 images with 21,944 bounding boxes across 1,180 classes.

Step 4: Automated Outlier Detection. We employed CLIP ViT-B/16 to extract image features and identify outliers within each class. Images with features more than 1.5 standard deviations from their class centroid were flagged and removed, reducing the dataset to 7,537 images with 17,513 bounding boxes.

Step 5: Manual Quality Control. Human reviewers examined remaining samples to correct or remove invalid bounding boxes and labels. To maintain meal composition integrity, we applied an all-or-none removal policy at the image level, resulting in 6,925 images with 14,349 bounding boxes from 513 users.

Step 6: Label Consolidation. The final step addressed label synonymity and inconsistency. We used GPT-4o to generate initial groupings of semantically similar labels, followed by manual refinement to create 218 final classes. This process maintained the natural class imbalance while ensuring semantic coherence.

3.3 Dataset Characteristics

The final FoodLogAthl-218 dataset exhibits several key characteristics that distinguish it from existing food image datasets:

Natural Class Imbalance. Unlike balanced web-sourced datasets, FoodLogAthl-218 reflects actual eating patterns with a heavy-tailed distribution. The most common classes include staple foods like rice and bread, while specialty items appear infrequently.

Temporal Structure. Each sample includes precise timestamps, enabling analysis of temporal patterns in food consumption and user behavior over time.

Multi-dish Images. Real meal photos often contain multiple dishes, with an average of 2.07 dishes per image, challenging single-dish classification models.

Casual Imaging Conditions. Images exhibit the natural variability of user-captured photos, including varied lighting, angles, and compositions that reflect real-world usage scenarios.

To quantify the increased difficulty compared to existing datasets, we computed distribution metrics using multiple feature extractors. Table 2 shows that FoodLogAthl-218 exhibits higher intra-class variation (Trace: 81.70 vs 71.21 for CLIP features) and lower cluster compactness (Silhouette: -0.021 vs 0.033) compared to UEC-Food 256, confirming greater classification difficulty.

4 Benchmark Tasks and Experimental Setup

We design three evaluation tasks to comprehensively assess model performance on practical food recognition challenges inherent in real-world deployment scenarios.

4.1 Task 1: Closed-Set Dish Classification

This task evaluates standard classification performance on individual dish crops extracted from the bounding boxes. Each model receives a cropped dish image and must predict the correct class from the 218 categories. We split the dataset with 70% for training, 15% for validation, and 15% for testing, ensuring temporal ordering is preserved to avoid data leakage.

Evaluation Metrics. We report Top-1, Top-3, and Top-5 recall (R@1, R@3, R@5) and macro-averaged F1 score to account for class imbalance.

4.2 Task 2: Continual User-Aware Adaptation

This task simulates realistic deployment where models encounter new user data over time and must adapt accordingly. We evaluate both zero-shot performance and continual adaptation capabilities along the temporal axis.

Experimental Setup. For each week in the dataset timeline, we evaluate model performance on that week's samples using only data from previous weeks for training/adaptation. We report both weekly performance and a weighted one-month moving average (±2 weeks), where weights are proportional to weekly data volume to account for varying sample sizes.

Adaptation Methods. We compare three approaches: (1) zero-shot inference without adaptation, (2) full fine-tuning on all previous user data, and (3) Low-Rank Adaptation (LoRA) fine-tuning for efficient parameter updates.

4.3 Task 3: Menu-Context-Aware Classification

Real meal images often contain multiple dishes, and knowledge of accompanying dishes can provide valuable context for classification. This task evaluates whether models can leverage menu-level information to improve individual dish recognition.

Table 1: Data Filtering Process and the Number of Images, Bounding Boxes (bboxes) and Classes.

Step	Operation	#images	(diff)	#bboxes	(diff)	#classes	(diff)
0	Raw data	11,460		61,414		5,889	
1	Remove samples w/o image	11,460	(0)	26,646	(-34,768)	3,167	(-2,722)
2	Remove samples w/o bounding box	11,285	(-175)	26,065	(-581)	3,071	(-96)
3	Remove classes w/ <10 samples	9,121	(-2,164)	21,944	(-4,121)	1,180	(-1,891)
4	Remove outliers based on image features	7,537	(-1,584)	17,513	(-4,431)	1,180	(0)
5	Manual review and remove invalid images	6,925	(-612)	14,349	(-3,164)	1,180	(0)
6	Merge similar classes and clean labels	6,925	(0)	14,349	(0)	218	(-982)

Table 2: Comparison of distribution metrics between FoodLogAthl-218 and UEC-Food 256 using feature extractors. Higher *Trace* and *Davies–Bouldin* indicate greater spread, while lower *Silhouette* and *Calinski–Harabasz* indicate less compact clusters.

	Trace (↑)			Silhouette (↓)			Davies–Bouldin (↑)			Calinski–Harabasz (↓)		
Dataset	**CLIP**	**EVA-CLIP**	**Swin**	**CLIP**	**EVA-CLIP**	**Swin**	**CLIP**	**EVA-CLIP**	**Swin**	**CLIP**	**EVA-CLIP**	**Swin**
UEC-Food 256	71.21	45.09	264.00	0.0335	0.0577	-0.0064	3.64	3.02	4.11	74.05	115.80	49.17
FoodLogAthl-218 (Ours)	**81.70**	**49.92**	**282.57**	**-0.0210**	**0.0036**	**-0.0288**	**4.52**	**3.68**	**4.51**	**27.51**	**51.51**	**21.29**

Table 3: Comparison of Datasets for Food Image Classification Tasks.

Year	Dataset Name	#classes	#samples	Cuisine Category	Data Source
2009	PFID [3]	101	4,545	Western	Fast Food Chains
2014	UEC-Food 256 [8]	256	25,088	International	**Web** + Original Database
2014	Food-101 [2]	101	101,000	Western	**Web** (foodspotting.com)
2015	UPMC Food-101 [13]	101	90,840	Western	**Web** (Google [6])
2016	UNICT-FD1200 [5]	1,200	4,754	International	Experiment Participants
2017	Vegfru [7]	25/292*	160,000	Vegetables, Fruits	**Web** (Google, ImageNet [4], etc.)
2019	FoodAI-756 [12]	152/756*	400,000	International	**Web** (Google, Bing, etc.)
2021	Food2K [11]	2,000	1,036,564	International	**Web** (Meituan [10])
2023	Food-500 Cap [9]	494	24,700	International	**Web** (Google, Bing)
2025	FoodLogAthl-218	218	14,349	International	**Dietary Management App**

* Hierarchically structured dataset, where n/m indicates m classes divided into n higher-level classes.

Methodology. For images containing multiple dishes, we generate free-form descriptions of all visible dishes using large multimodal models. These descriptions are embedded using intfloat/e5-large-v2, and classification is performed by selecting the label with the highest cosine similarity to the dish description embedding.

Evaluation. We compare performance with and without menu context using the same metrics as Task 1, analyzing both improvements and trade-offs in precision versus recall.

4.4 Models and Implementation

We evaluate a comprehensive set of models spanning different architectures and capabilities:

Large Multimodal Models: GPT-4o, Qwen2.5-VL-7B-Instruct, Qwen2-VL-7B-Instruct, Qwen2-VL-3B-Instruct, LLaVA-OneVision-Qwen2-7B-OV, LLaVA-v1.6-34B, LLaVA-v1.5-7B, and LLaVA-v1.6-Mistral-7B.

Vision-Language Models: CLIP ViT-L/14 and EVA-CLIP ViT-g/14 for embedding-based classification.

All experiments use consistent data splits and evaluation protocols. For LMM-based inference, we employ systematic prompting strategies and evaluate using the lmms-eval framework for reproducibility.

5 Results

5.1 Closed-Set Classification Performance

Table 4 presents comprehensive results for Task 1 across all evaluated models. The challenging nature of FoodLogAthl-218 is immediately apparent, with the best-performing model (GPT-4o) achieving only 42.4% top-1 accuracy and 0.373 F1 score.

Among the large multimodal models, GPT-4o significantly outperforms others across all metrics, followed by Qwen2.5-VL-7B-Instruct with 36.8% R@1. Notably, vision-language models like CLIP ViT-L/14 achieve competitive top-5 performance (60.0%) while struggling with top-1 accuracy (25.7%), suggesting that correct labels often appear in the top predictions but disambiguation remains challenging.

The substantial performance gap compared to existing benchmarks (e.g., >90% accuracy on Food-101 [2]) confirms that FoodLogAthl-218 presents significantly greater challenges, reflecting real-world complexity absent in web-sourced datasets.

5.2 Continual User-Aware Adaptation

The temporal adaptation results demonstrate striking benefits from continual learning approaches. Zero-shot models initially achieve approximately 20% accuracy but show minimal improvement over time. In contrast, models with LoRA fine-tuning exhibit substantial performance gains, reaching over 50% accuracy with 18 months of data exposure—a gain exceeding 30 percentage points.

Full fine-tuning shows similar improvement patterns but requires significantly more computational resources. LoRA provides an efficient alternative that achieves comparable adaptation performance while maintaining practical deployment feasibility.

Personalization Analysis. User-specific adaptation reveals even more dramatic improvements. For a heavy-logging user (User A), personalized LoRA fine-tuning increases accuracy by approximately 50 percentage points compared to zero-shot performance, eventually surpassing the all-users adaptation model. This suggests that individual eating patterns contain learnable structure that significantly benefits model performance.

Temporal Pattern Analysis. Detailed examination of the adaptation curves reveals distinct learning phases that align with practical deployment considerations. During the initial 4-week period, adaptation models show rapid improvement primarily on common staple foods (rice, bread, vegetables), achieving 15-20% accuracy gains. Subsequently, a slower learning phase emerges where models begin recognizing user-specific preferences and cooking styles, particularly for culturally-specific dishes and personal preparation variations. This pattern suggests that practical deployment strategies should prioritize initial adaptation on frequently consumed foods while gradually incorporating personalized recognition capabilities.

The temporal analysis also reveals seasonal eating patterns that impact model adaptation. Users exhibit increased consumption of specific foods during holidays and seasonal periods, creating natural distribution shifts that models must handle. For instance, traditional Japanese dishes show increased frequency during year-end periods, while lighter meals peak during summer months. Models adapted over longer periods (6+ months) demonstrate improved robustness to these seasonal variations, suggesting that extended adaptation periods provide benefits beyond simple volume accumulation.

User Behavior Pattern Analysis. Detailed examination of individual user trajectories reveals distinct behavioral archetypes that influence model adaptation efficiency. Heavy users (>10 logs/week) enable rapid model convergence within 3-4 weeks, achieving 35-40% accuracy gains, while occasional users (<3 logs/week) require 8-12 weeks to reach similar improvement levels. Interestingly, users with diverse dietary patterns (>50 distinct dishes logged) show more stable adaptation curves compared to users with repetitive eating habits, suggesting that dietary diversity acts as a natural regularization mechanism against overfitting.

Analysis of error patterns reveals that temporal adaptation primarily improves recognition of culturally-specific and personally-prepared dishes, with accuracy gains of 45-60% on these categories compared to 15-25% gains on commercially-prepared foods. This suggests that continual learning particularly benefits from exposure to consistent preparation styles and cultural food contexts that are underrepresented in web-sourced training data.

Further investigation into user clustering patterns based on dietary preferences reveals distinct behavioral profiles that significantly impact adaptation effectiveness. Health-conscious users who predominantly consume fresh, minimally processed foods (23% of our user base) demonstrate rapid model convergence within 2-3 weeks, achieving mean accuracy improvements of 42% due to consistent lighting conditions and ingredient visibility in their photos. In contrast, convenience-food-oriented users (31% of users) present greater recognition challenges due to packaged foods and varied presentation styles, requiring 6-8 weeks for comparable adaptation gains.

Geographic and cultural factors also emerge as critical determinants of adaptation efficiency. Users from traditional Japanese households (identified through predominant consumption of rice-based meals, miso soup, and pickled vegetables) show remarkably consistent imaging patterns and preparation styles, enabling models to achieve 38% accuracy improvements within 4 weeks. Western cuisine-oriented users exhibit more varied food presentation and preparation methods, resulting in slower but ultimately higher adaptation ceilings, reaching 45-50% improvements after 12-week exposure periods.

5.3 Menu-Context-Aware Classification

Table 5 shows mixed results for incorporating menu-level context. While menu-aware classification improves F1 score from 0.159 to 0.230 and Precision@1 from 0.102 to 0.179, it reduces Recall@1 from 0.366 to 0.324, indicating a precision-recall trade-off.

Detailed error analysis reveals that menu context corrects 687 classification errors while introducing 1,222 new errors. This suggests that while contextual information can provide valuable disambiguation cues, it may also introduce biases that lead models astray when context is misleading or incomplete.

Context Effectiveness Analysis. The effectiveness of menu context varies significantly across food categories and meal compositions. Context proves most beneficial for visually similar dishes within the same cuisine family, such as distinguishing between different types of Japanese rice bowls (donburi) or Chinese stir-fried dishes. In these cases, accompanying dishes provide strong cultural and preparation method cues that improve classification accuracy by 25-35%.

Conversely, context introduces errors primarily in cross-cultural meals and fusion dishes where accompanying items may provide misleading signals. For instance, when traditional Japanese dishes appear alongside Western items, the context embedding may incorrectly bias classification toward fusion variants or entirely different categories.

Embedding Quality and Similarity Thresholds. Analysis of the cosine similarity distributions reveals that effective context application requires careful threshold selection. High-confidence context matches (cosine similarity >0.7) show 68% correction accuracy, while low-confidence matches (<0.4) introduce errors in 71% of cases. This suggests that adaptive thresholding based on context confidence could significantly improve the precision-recall balance.

The e5-large-v2 embeddings capture semantic relationships effectively for common food combinations but struggle with regional variations and unconventional pairings. Alternative embedding

strategies, such as food-domain-specific models or multi-modal context representations, may provide improved contextual reasoning capabilities for complex meal compositions.

5.4 Dataset Difficulty Analysis

To quantify the increased difficulty of FoodLogAthl-218, we compared feature-space distributions with UEC-Food 256 using multiple extractors. Our dataset exhibits consistently higher intra-class variation (CLIP Trace: 81.70 vs 71.21) and lower cluster compactness (Silhouette: -0.021 vs 0.033), confirming greater classification difficulty across different feature spaces.

This increased difficulty stems from several factors: natural class imbalance, varied imaging conditions, multi-dish compositions, and the inherent noise in user-generated content. Unlike carefully curated web datasets, FoodLogAthl-218 preserves the challenges that models must handle in practical deployment.

5.5 Computational Analysis and Deployment Considerations

The practical deployment of food recognition systems requires careful consideration of computational resources and latency constraints. Our extensive evaluation provides insights into the computational trade-offs across different model architectures and adaptation strategies.

Inference Efficiency. Among large multimodal models, GPT-4o achieves the highest accuracy but requires cloud-based inference with significant latency (2-4 seconds per image) and cost implications. In contrast, the 7B parameter Qwen models provide reasonable accuracy (36.8% R@1) with local deployment capability, achieving inference times of 0.3-0.5 seconds on modern mobile hardware.

Vision-language models like CLIP offer the most efficient deployment option, with inference times under 100ms while maintaining competitive top-5 performance (60.0%). This suggests a viable path for real-time mobile applications where top-k suggestions may be sufficient for user interaction.

Adaptation Scalability. The LoRA adaptation approach demonstrates superior computational efficiency compared to full fine-tuning, requiring only 2-5% of the parameter updates while achieving comparable performance gains. For a deployment serving 1000+ users, LoRA enables personalized adaptation with manageable storage overhead (8-15MB per user model) and training time (5-10 minutes per user per month).

Memory requirements for temporal adaptation scale linearly with the number of active users and adaptation history length. Our experiments suggest that maintaining 3-month adaptation windows provides optimal performance-efficiency balance, requiring approximately 50-100MB storage per active user for historical embeddings and adaptation checkpoints.

6 Conclusion

We present FoodLogAthl-218, the first large-scale food image dataset constructed directly from real dietary management app usage, addressing a critical gap between laboratory benchmarks and practical deployment scenarios. Our dataset contains 6,925 images with 14,349 bounding boxes across 218 classes, preserving the natural challenges of real-world food recognition including class imbalance, casual imaging conditions, multi-dish compositions, and temporal patterns.

Table 4: Classification performance on FoodLogAthl-218.

Model	F1	R@1	R@3	R@5
GPT-4o	**0.3732**	**0.4243**	**0.4991**	0.5221
Qwen2.5-VL-7B-Instruct	0.2349	0.3684	0.4609	0.4859
Qwen2-VL-7B-Instruct	0.1594	0.3656	0.4590	0.4781
Qwen2-VL-3B-Instruct	0.1386	0.2663	0.3250	0.3517
llava-onevision-qwen2-7b-ov	0.1546	0.3629	0.4497	0.4639
EVA-CLIP (ViT-g)	0.2465	0.2654	0.4897	0.5944
CLIP (ViT-L/14)	0.2393	0.2567	0.4885	**0.6000**
llava-v1.6-34b	0.1354	0.2432	0.3779	0.4063
llava-v1.5-7b	0.0991	0.2140	0.2721	0.2973
llava-v1.6-mistral-7b	0.0944	0.1932	0.2595	0.2831

Table 5: F1 score, Precision@1 (P@1), and Recall@K (R@1, R@3, R@5) for Qwen2.5-VL-7B-Instruct on the description-guided classification task.

	F1	P@1	R@1	R@3	R@5
w/o description	0.1594	0.1019	**0.3656**	**0.4590**	**0.4781**
w/ description	**0.2304**	**0.1788**	0.3240	0.4032	0.4336

Through comprehensive evaluation across three benchmark tasks, we demonstrate that FoodLogAthl-218 presents significantly greater challenges than existing datasets, with state-of-the-art models achieving substantially lower performance compared to web-sourced benchmarks. Our experiments reveal several key insights: (1) continual adaptation provides substantial benefits with over 30 percentage point improvements through temporal learning, (2) personalization yields even greater gains for individual users, and (3) menu-context-aware classification shows promising precision improvements but with recall trade-offs.

The reproducible data curation pipeline, combining automated outlier detection with manual quality control, provides a framework for constructing similar real-world datasets in other domains. Our comprehensive baselines with large multimodal models and vision-language models establish benchmarks for future research.

FoodLogAthl-218 enables research on practical food recognition challenges including continual learning, personalization, and context-aware classification that are critical for real-world deployment but underexplored in existing benchmarks. We release the dataset publicly to facilitate research toward more robust and practical food recognition systems.

Limitations. Our dataset reflects the user demographics and regional preferences of the FoodLog Athl app, which may limit generalizability to other populations. Additionally, the manual curation process, while necessary for quality control, may introduce annotation biases. Future work should explore expanding the dataset to include more diverse populations and developing automated quality control methods.

Future Work. Several research directions emerge from our work: developing more effective continual adaptation strategies for vision models, investigating personalization techniques that

balance individual adaptation with general model capabilities, and exploring improved context-aware architectures that can better leverage multi-dish information.

7 Acknowledgements

We thank the users of the FoodLog Athl application for their data contributions and consent to this research. We also acknowledge the manual reviewers who contributed to the dataset curation process and the computational resources provided for model training and evaluation.

References

[1] Lukas Bossard, Matthieu Guillaumin, and Luc Van Gool. 2014. Food-101-mining discriminative components with random forests. In *ECCV*.

[2] Lukas Bossard, Matthieu Guillaumin, and Luc Van Gool. 2014. Food-101–mining discriminative components with random forests. In *ECCV*.

[3] Mei Chen, Kapil Dhingra, Wen Wu, Lei Yang, Rahul Sukthankar, and Jie Yang. 2009. PFID: Pittsburgh fast-food image dataset. In *ICIP*.

[4] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. 2009. Imagenet: A large-scale hierarchical image database. In *CVPR*.

[5] Giovanni Maria Farinella, Dario Allegra, Marco Moltisanti, Filippo Stanco, and Sebastiano Battiato. 2016. Retrieval and classification of food images. *Computers in biology and medicine* 77 (2016), 23–39.

[6] Google LLC. 2024. Google Search. https://www.google.com/. (Accessed on 16/03/2024).

[7] Saihui Hou, Yushan Feng, and Zilei Wang. 2017. Vegfru: A domain-specific dataset for fine-grained visual categorization. In *ICCV*.

[8] Y. Kawano and K. Yanai. 2014. Automatic Expansion of a Food Image Dataset Leveraging Existing Categories with Domain Adaptation. In *ECCV Workshops (TASK-CV)*.

[9] Zheng Ma, Mianzhi Pan, Wenhan Wu, Kanzhi Cheng, Jianbing Zhang, Shujian Huang, and Jiajun Chen. 2023. Food-500 Cap: A Fine-Grained Food Caption Benchmark for Evaluating Vision-Language Models. In *ACMMM*.

[10] Meituan. 2024. Meituan. https://www.meituan.com/. (Accessed on 16/03/2024).

[11] Weiqing Min, Zhiling Wang, Yuxin Liu, Mengjiang Luo, Liping Kang, Xiaoming Wei, Xiaolin Wei, and Shuqiang Jiang. 2023. Large scale visual food recognition. *TPAMI* (2023).

[12] Doyen Sahoo, Wang Hao, Shu Ke, Wu Xiongwei, Hung Le, Palakorn Achananuparp, Ee-Peng Lim, and Steven CH Hoi. 2019. FoodAI: Food image recognition via deep learning for smart food logging. In *KDD*.

[13] Xin Wang, Devinder Kumar, Nicolas Thome, Matthieu Cord, and Frederic Precioso. 2015. Recipe recognition with large multimodal food dataset. In *ICMEW*.